



Bank Transaction Fraud Detection

POWER BI DASHBOARD PRESENTATION

Project Overview

- ▶ This project utilizes a bank transaction dataset to identify patterns of fraudulent activities.
- ▶ It focuses on building a data-driven Power BI dashboard that empowers analysts to detect fraud effectively using interactive visuals and DAX calculations.

Objectives

- ▶ Detect fraudulent transactions using transaction and balance patterns.
- ▶ Visualize key fraud indicators through Power BI.
- ▶ Provide actionable insights using DAX measures and filters.

Dataset Summary

- ▶ Fields: step, type, amount, nameOrig, nameDest, oldbalanceOrg, newbalanceOrig, oldbalanceDest, newbalanceDest, isFraud, isFlaggedFraud
- ▶ Size: ~1 million transactions
- ▶ Source: Simulated bank transaction logs

Key Problems & Solutions

- ▶ Problem: Which transaction types are more fraud-prone?
- ▶ Solution: Bar chart with fraud % by type

- ▶ Problem: How to spot suspicious transactions?
- ▶ Solution: Scatter plots for amount vs balance drops

- ▶ Problem: Who are frequent fraud actors?
- ▶ Solution: Top 10 origin/destination accounts by fraud count

- ▶ Problem: Are false positives common?
- ▶ Solution: Use DAX to compare is Fraud vs is Flagged Fraud

Power BI Dashboard Elements

Visuals:

- ▶ Bar Chart: Fraud % by Transaction Type
- ▶ Line Chart: Fraud Cases Over Time (step)
- ▶ KPI Cards: Total Frauds, Fraud %, Avg Fraud Amount
- ▶ Table: Top Fraudulent Accounts
- ▶ Scatter Plot: Balance Discrepancy vs. Amount

Slicers:

- ▶ Transaction Type, Time Step (step), Fraud Indicator (is Fraud), Amount Range

Key DAX Measures

- ▶ Total Frauds = `CALCULATE(COUNTROWS('Dataset'), 'Dataset'[isFraud] = 1)`
- ▶ Fraud % = `DIVIDE([Total Frauds], COUNTROWS('Dataset'))`
- ▶ Avg Fraud Amount = `CALCULATE(AVERAGE('Dataset'[amount]), 'Dataset'[isFraud] = 1)`
- ▶ False Positives = `CALCULATE(COUNTROWS('Dataset'), 'Dataset'[isFlaggedFraud] = 1 && 'Dataset'[isFraud] = 0)`
- ▶ Balance Drop = `'Dataset'[oldbalanceOrg] - 'Dataset'[newbalanceOrig]`

Conclusion

This Power BI dashboard serves as an effective fraud monitoring tool. It reveals fraudulent patterns, accounts at risk, and potential false positives, aiding banking institutions in minimizing fraud-related losses.



THANKING YOU